A Time-Block Holdout Strategy for Evaluating AR(2) Temporal Pooling in a Spatial Skew-t Model

Research Technical Note: `time_block_strategy`

Spatial Skew-t Simulation Study

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Abstract

A simulation study fits a spatial skew-t model with and without AR(2) temporal structure on three latent knot-level processes (precision τ, skewness z, and knot location w), and scores predictions with the Brier score. The AR(2) variants fail to dominate their independent-in-time counterparts, and in one setting score strictly worse. This note formalizes why: the Brier score is a strictly proper score for the *marginal* exceedance probability, while the existing holdout removes spatial *sites* and retains all time points. Under that design the AR(2) prior and the i.i.d. prior induce the same marginal predictive distribution at an unobserved site, so a marginal score cannot separate them and the simpler model wins the bias–variance trade-off. We prove that the conditional h-step AR(2) forecast converges geometrically to the stationary marginal at rate governed by the companion-matrix spectral radius, define an effective memory horizon, and use it to design a time-block holdout that forecasts forward from the posterior of the seam state. We give the forecasting algorithm, the latent-scale coupling of τ and z, and a lead-time scoring protocol based on multivariate proper scores. The result is an evaluation framework under which AR(2) temporal pooling can genuinely be tested rather than structurally nullified.

1 Introduction

A spatial skew-t model for extreme environmental fields can be extended with temporal dependence by placing autoregressive priors on its latent knot-level processes. The natural question is whether this temporal pooling improves predictive performance for extremes. A simulation study addressed this by fitting eight method variants — including two that activate AR(2) structure on all three latent processes — and scoring held-out predictions with the Brier score across a grid of high threshold quantiles.

The outcome was a null result. The AR(2) variants did not dominate the independent-in-time skew-t and t variants; in the setting with five knots the AR(2) model scored *worse* than its own non-temporal sibling at every threshold, with relative Brier inflation reaching 1.16 in the far upper tail. Taken at face value this suggests temporal pooling is unhelpful. We argue instead that the result is an artifact of the *interaction between the scoring rule and the holdout design*, and that the experiment as constructed cannot answer the question it was built to ask.

This note has three goals. First, in Section **??** we formalize the model, the AR(2) latent processes, and the family of scoring rules. Second, in Section **??** we prove that a marginal score under a spatial-only holdout is structurally blind to temporal dependence, and we quantify the AR(2) memory horizon through the companion-matrix spectral radius. Third, in Section **??** we derive a corrected time-block holdout: its forecasting algorithm, the seam-state conditioning that preserves the temporal signal, the latent-scale transform order, and a lead-time scoring protocol. Section **??** states the expected outcome and open issues.

2 Model Setup and Definitions

2.1 The spatial skew-t observation model

Let $\mathcal{S} \subset \mathbb{R}^2$ be a study region and let $t \in \{1, \dots, n_t\}$ index time. Observations are made at sites $s \in \mathcal{S}$ and times t.

Definition 1 (Observation model)**.** For site s and time t,

$$y(s,t) = x(s,t)^\top \beta + \lambda\, z_{g(s,t),\,t} + \varepsilon(s,t), \tag{1}$$

where $x(s,t) \in \mathbb{R}^p$ are covariates with coefficient vector $\beta \in \mathbb{R}^p$; λ is a skewness loading; $\varepsilon(\cdot,t)$ is a mean-zero spatial Gaussian field with $\mathrm{Cov}\{\varepsilon(s,t), \varepsilon(s',t)\} = C(s,s')\,\tau^{-1}_{g(s,t),\,t}$; and C is a correlation function with range ρ, smoothness ν, and geometric-anisotropy parameter γ.

The model carries K spatial *knots*. For each knot $k \in \{1, \dots, K\}$ and time t there are three latent quantities: a precision $\tau_{k,t} > 0$, a skewness term $z_{k,t} \geq 0$, and a knot location $w_{k,t} \in \mathbb{R}^2$. The function $g(s,t)$ returns the index of the knot nearest to s at time t, i.e. the Voronoi membership

$$g(s,t) = \operatorname*{arg\,min}_{k \in \{1,\dots,K\}} \big\| s - w_{k,t} \big\|. \tag{2}$$

Assumption 1 (Conditional independence of the spatial nugget)**.** The spatial field $\varepsilon(\cdot,t)$ in (**??**) is drawn independently across t. All cross-time dependence in y is therefore transmitted exclusively through the latent processes $\{\tau_{k,t}, z_{k,t}, w_{k,t}\}$.

Assumption **??** is a property of the data-generating mechanism, not a modelling choice; it localizes the entire temporal signal in the latent processes and is the hinge of every argument that follows.

2.2 Latent AR(2) processes on the Gaussian scale

Each latent process is generated on a standardized Gaussian scale and then mapped to its target marginal by a copula transform. Write X_t generically for any of the latent Gaussian processes $\tau^\star_{k,\cdot}$, $z^\star_{k,\cdot}$, or $w^\star_{k,\cdot}$.

Definition 2 (AR(2) latent process)**.** The latent Gaussian process obeys

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \tag{3}$$

with coefficients $\phi = (\phi_1, \phi_2)$ and innovation variance σ^2.

Definition 3 (Stationarity triangle)**.** The recursion (**??**) is stationary if and only if

$$|\phi_2| < 1, \qquad \phi_1 + \phi_2 < 1, \qquad \phi_2 - \phi_1 < 1. \tag{4}$$

We normalize each latent process to unit marginal variance, $\gamma_0 := \mathrm{Var}(X_t) = 1$. The Yule–Walker relations then fix the lag-one and lag-two autocovariances and the innovation variance:

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2}, \qquad \gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0, \qquad \sigma^2 = \gamma_0 - \phi_1 \gamma_1 - \phi_2 \gamma_2. \tag{5}$$

Definition 4 (Companion form)**.** Stacking $\boldsymbol{X}_t = (X_t, X_{t-1})^\top$, the recursion (**??**) becomes the first-order vector system

$$\boldsymbol{X}_t = F\,\boldsymbol{X}_{t-1} + \boldsymbol{e}_t, \qquad F = \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix}, \qquad \boldsymbol{e}_t = \begin{pmatrix} \xi_t \\ 0 \end{pmatrix}, \tag{6}$$

with innovation covariance $Q = \mathrm{diag}(\sigma^2, 0)$. Stationarity (**??**) is equivalent to the spectral radius condition $\rho_* := \rho(F) < 1$, where $\rho(F)$ is the modulus of the largest eigenvalue of F.

Definition 5 (Stationary state distribution)**.** The stationary distribution of the companion state is $\boldsymbol{X}_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$, where Σ solves the discrete Lyapunov equation

$$\Sigma = F\,\Sigma\,F^\top + Q, \qquad \Sigma = \begin{pmatrix} \gamma_0 & \gamma_1 \\ \gamma_1 & \gamma_0 \end{pmatrix} = \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix}. \tag{7}$$

Initializing a simulated path by drawing (X_1, X_2) from $\mathcal{N}(\mathbf{0}, \Sigma)$ yields an *exactly* stationary series with no burn-in. This is the construction used by the study's generator: a manual AR(p) recursion with an exact stationary initial state.

2.3 Copula transforms to observable scale

The latent Gaussian processes are mapped to their target marginals by monotone copula transforms, applied *after* the AR(2) recursion:

$$\tau_{k,t} = G^{-1}_{\alpha,\beta}\big(\Phi(\tau^\star_{k,t})\big), \tag{8}$$

$$z_{k,t} = H^{-1}_{\sigma_z}\big(\Phi(z^\star_{k,t})\big), \qquad \sigma_z = \tau_{k,t}^{-1/2}, \tag{9}$$

$$w_{k,t} = \Psi^{-1}\big(\Phi(w^\star_{k,t})\big), \tag{10}$$

where Φ is the standard normal CDF, $G^{-1}_{\alpha,\beta}$ is a gamma quantile function, $H^{-1}_{\sigma_z}$ is a half-normal quantile function with scale σ_z, and Ψ^{-1} is a scaled inverse-probit map onto the spatial domain. Equation (**??**) records a structural coupling: the skewness scale σ_z is itself a function of the precision $\tau_{k,t}$.

2.4 Scoring rules

Let F denote a predictive distribution and y the realized outcome.

Definition 6 (Brier score)**.** For threshold c, with predicted exceedance probability $\hat{p}(s,t;c) = \mathbb{P}_F\{Y(s,t) > c\}$ and indicator $\mathbf{1}\{y(s,t) > c\}$, the Brier score is the cell-averaged quadratic loss

$$\mathrm{BS}(c) = \frac{1}{|\mathcal{V}|} \sum_{(s,t)\in\mathcal{V}} \big(\mathbf{1}\{y(s,t) > c\} - \hat{p}(s,t;c)\big)^2, \tag{11}$$

where $\mathcal{V}$ is the set of held-out (s,t) cells.

Definition 7 (Energy score)**.** For a vector outcome $\boldsymbol{y} \in \mathbb{R}^d$ and predictive F on $\mathbb{R}^d$,

$$\mathrm{ES}(F, \boldsymbol{y}) = \mathbb{E}_F\|\boldsymbol{X} - \boldsymbol{y}\| - \tfrac{1}{2}\,\mathbb{E}_F\|\boldsymbol{X} - \boldsymbol{X}'\|, \tag{12}$$

where $\boldsymbol{X}, \boldsymbol{X}' \overset{\text{iid}}{\sim} F$. The energy score is a strictly proper score for the *joint* law of $\boldsymbol{y}$. Its univariate case $d = 1$ is the continuous ranked probability score (CRPS).

Definition 8 (Variogram score of order p)**.** For $\boldsymbol{y} \in \mathbb{R}^d$ and weights $w_{ij} \geq 0$,

$$\mathrm{VS}_p(F, \boldsymbol{y}) = \sum_{i=1}^{d} \sum_{j=1}^{d} w_{ij} \Big(|y_i - y_j|^p - \mathbb{E}_F |X_i - X_j|^p \Big)^2. \tag{13}$$

Indexing i, j by time lag makes VS_p sensitive to the autocorrelation structure of the predictive.

3 Why a Marginal Score Cannot See Temporal Structure

This section establishes the diagnosis. The argument has three steps: the Brier score depends on the predictive only through marginals (Proposition **??**); a spatial-only holdout makes the AR(2) and i.i.d. models share those marginals (Proposition **??**); and the AR(2) advantage that *does* exist lives in a forecast horizon that the holdout never exercises (Propositions **??**–**??**).

3.1 The Brier score is a marginal functional

Proposition 1 (Brier insensitivity to dependence)**.** *The Brier score* (**??**) *depends on the predictive distribution F only through the collection of univariate marginal exceedance probabilities $\{\hat{p}(s, t; c) : (s, t) \in \mathcal{V}\}$. Consequently, if two predictive models F_A and F_B induce identical marginals at every held-out cell — $\hat{p}_A(s, t; c) = \hat{p}_B(s, t; c)$ for all $(s, t) \in \mathcal{V}$ and all c — then they receive identical Brier scores, regardless of any difference in their dependence structure.*

Proof. Each summand of (**??**) is a function of the pair $\big(\mathbf{1}\{y(s,t) > c\}, \hat{p}(s,t;c)\big)$. The indicator is a property of the data, not of F. The only term through which F enters is the scalar $\hat{p}(s,t;c) = \mathbb{P}_F\{Y(s,t) > c\}$, which is a functional of the univariate marginal of F at cell (s,t). The sum contains no term coupling two distinct cells. Hence $\mathrm{BS}(c)$ is a function of the marginals alone, and any two predictives agreeing on all marginals produce the same value. □ □

Remark 1. Proposition **??** is not a defect of the Brier score; it is its design. The Brier score is a strictly proper score for a *marginal* binary event. Cross-cell dependence is simply outside the quantity it was built to assess. The energy and variogram scores of Definitions **??**–**??** are the proper scores that *do* interrogate dependence.

3.2 Spatial-only holdout collapses the comparison

Consider the holdout used by the study: a subset of spatial sites $\mathcal{S}_{\mathrm{val}}$ is withheld, while *every* time point is observed at the training sites. Prediction targets are the cells $\{(s,t) : s \in \mathcal{S}_{\mathrm{val}},\ t = 1, \ldots, n_t\}$.

Proposition 2 (Structural collapse under spatial-only holdout)**.** *Under Assumption* **??** *and a spatial-only holdout, the predictive marginal at an unobserved site $s^\star$ and time t is*

$$\mathbb{P}\{Y(s^\star, t) > c\} = \mathbb{E}_{\tau, z, w}\big[\mathbb{P}\big\{Y(s^\star, t) > c \,\big|\, \tau_{\cdot,t}, z_{\cdot,t}, w_{\cdot,t}\big\}\big], \tag{14}$$

where the expectation is over the time-t marginal *law of the latent processes. The AR(2) prior and the i.i.d. prior, when both are normalized to the same stationary marginal $\mathcal{N}(0,1)$ on the latent scale, induce the same law for $(\tau_{\cdot,t}, z_{\cdot,t}, w_{\cdot,t})$ at each fixed t, hence the same marginal* (**??**)*, hence — by Proposition* **??** *— the same Brier score in expectation.*

Proof. By Assumption **??** the spatial nugget at time t is independent of all other times, so conditioning on the time-t latent slice $(\tau_{\cdot,t}, z_{\cdot,t}, w_{\cdot,t})$ renders $Y(s^\star, t)$ independent of every other time. The exceedance probability at cell $(s^\star, t)$ therefore depends on the latent processes only through their time-t values, giving (**??**).

The AR(2) prior of Definition **??** and an i.i.d. prior differ only in the *joint* law across t; both are constructed with the same unit-variance stationary marginal $X_t \sim \mathcal{N}(0, 1)$ on the latent scale (the normalization (**??**)). The copula transforms (**??**)–(**??**) are applied marginally and time-pointwise, so equal latent marginals yield equal marginals for $(\tau_{\cdot,t}, z_{\cdot,t}, w_{\cdot,t})$. The right-hand side of (**??**) is thus identical for the two priors. Proposition **??** converts the equal marginals into equal Brier scores. □ □

Remark 2. Proposition **??** explains the empirical observation. If the AR(2) and i.i.d. models are scored on a quantity for which they are exchangeable, the comparison reduces to a bias–variance contest. The AR(2) model carries six extra parameters — a pair (ϕ_1, ϕ_2) for each of τ, z, w — whose posterior uncertainty inflates the predictive variance without changing the scored marginal. The simpler model therefore wins, and the AR(2) penalty grows in the upper tail where estimation variance is largest. This is a property of the experiment, not of temporal pooling.

3.3 Where the AR(2) advantage actually lives

The AR(2) advantage is a statement about *conditional* forecasts. We make it precise.

Proposition 3 (Conditional h-step forecast)**.** *Let T_o be the last observed time and $\boldsymbol{X}_{T_o}$ the companion state* (**??**)*. For a stationary AR(2),*

$$\mathbb{E}[\boldsymbol{X}_{T_o+h} \mid \boldsymbol{X}_{T_o}] = F^h \boldsymbol{X}_{T_o}, \qquad \mathrm{Var}[\boldsymbol{X}_{T_o+h} \mid \boldsymbol{X}_{T_o}] = \Sigma_h := \sum_{j=0}^{h-1} F^j Q\, (F^j)^\top. \tag{15}$$

As $h \to \infty$, $F^h \to \mathbf{0}$ and $\Sigma_h \to \Sigma$, the stationary covariance of Definition **??***.*

Proof. Iterating (**??**) forward h steps gives $\boldsymbol{X}_{T_o+h} = F^h \boldsymbol{X}_{T_o} + \sum_{j=0}^{h-1} F^j \boldsymbol{e}_{T_o+h-j}$. The innovations $\boldsymbol{e}_\cdot$ are mean-zero and independent of $\boldsymbol{X}_{T_o}$, which yields the conditional mean and variance in (**??**). Stationarity gives $\rho_* = \rho(F) < 1$ (Definition **??**), so $F^h \to \mathbf{0}$ geometrically; the partial sum Σ_h is monotone increasing in the positive semidefinite order and converges to the solution Σ of the Lyapunov equation (**??**). □ □

Corollary 1 (The stationary distribution is the infinite-horizon limit)**.** *Drawing the forecast origin from the stationary distribution $\mathcal{N}(\mathbf{0}, \Sigma)$ is equivalent to the limit $h \to \infty$ of* (**??**)*. It discards the conditioning information in $\boldsymbol{X}_{T_o}$ and reproduces, for every h, the i.i.d. model's marginal predictive.*

Proof. Setting the mean term $F^h \boldsymbol{X}_{T_o}$ to zero and the variance to Σ in (**??**) is exactly the result of replacing $\boldsymbol{X}_{T_o}$ by a draw from $\mathcal{N}(\mathbf{0}, \Sigma)$ and recursing: the forecast law becomes $\mathcal{N}(\mathbf{0}, \Sigma)$ for all h, independent of h. That is the time-t stationary marginal, which by Proposition **??** is the i.i.d. predictive. □ □

Corollary **??** is the central implementation warning: initializing a forecast from the stationary distribution *is* the act of erasing the AR(2) signal. The forecast must instead be conditioned on the posterior of the seam state $\boldsymbol{X}_{T_o}$.

Proposition 4 (Effective memory horizon)**.** *The discrepancy between the conditional forecast* (**??**) *and the stationary marginal decays geometrically:*

$$\big\|\mathbb{E}[\boldsymbol{X}_{T_o+h} \mid \boldsymbol{X}_{T_o}]\big\| \le \|F^h\| \, \|\boldsymbol{X}_{T_o}\| = O\big(\rho_*^{\,h}\big), \qquad \|\Sigma - \Sigma_h\| = O\big(\rho_*^{\,2h}\big). \tag{16}$$

Hence for a tolerance $\epsilon > 0$ there is an effective memory horizon

$$h^\star(\epsilon) = \left\lceil \frac{\log \epsilon}{\log \rho_*} \right\rceil, \tag{17}$$

beyond which the AR(2) and i.i.d. predictives are indistinguishable to within ϵ.

Proof. Since $\rho_* < 1$, the spectral radius formula gives $\|F^h\| = O(\rho_*^{\,h})$ up to a polynomial factor absorbed into the constant; the mean bound in (**??**) follows. For the variance, $\Sigma - \Sigma_h = \sum_{j \ge h} F^j Q (F^j)^\top$, whose norm is bounded by a geometric tail of ratio ρ_*^2, giving $O(\rho_*^{\,2h})$. Setting the mean bound equal to ϵ and solving for h yields (**??**). □ □

Remark 3. Proposition **??** converts a modelling intuition into a design parameter. For coefficients of the order $\phi = (0.6, -0.3)$ the spectral radius is moderate and $h^\star$ is a single-digit number of steps. A holdout that scores predictions only beyond $h^\star$ measures nothing but the stationary marginal; a holdout that never forecasts past T_o at all — the spatial-only design — exercises the conditional forecast for exactly zero steps. Either way the AR(2) signal is invisible by construction.

4 A Time-Block Holdout That Exercises Temporal Pooling

The diagnosis dictates the cure: hold out *time blocks* and forecast forward from the posterior of the seam state, scoring with a rule that credits joint structure and resolving the result by lead time.

4.1 Holdout geometry

Definition 9 (Time-block holdout)**.** Choose B non-overlapping blocks indexed $b = 1, \dots, B$, each of length H with $H \lesssim h^\star$ from (**??**) (a practical range is $H \in [5, 15]$). Block b has seam time $T_o^{(b)}$ and prediction targets $\{(s,t) : s \in \mathcal{S},\ t = T_o^{(b)} + 1, \dots, T_o^{(b)} + H\}$. Blocks are spread across the record so adjacent blocks are approximately independent.

Choosing H near $h^\star$ keeps every scored lead time inside the window where the conditional forecast still differs from the stationary marginal (Proposition **??**).

4.2 Forecasting algorithm

Forecast pipeline at a glance. For each held-out block b with seam $T_o = T_o^{(b)}$ and horizon H, the predictive draw $\hat{y}^{(m)}(s, T_o + h)$ at lead h is built in *five* stages, once per MCMC sample $m = 1, \dots, M$.

(1) *Fit on the contiguous prefix.* The training data $y(\cdot, t \le T_o)$ feeds the MCMC, which returns M joint posterior draws $\theta^{(m)}$ together with imputed observable latent trajectories $\tau_{k,\cdot}^{(m)}$, $z_{k,\cdot}^{(m)}$, $w_{k,\cdot}^{(m)}$ for every $t \le T_o$. All sites are observed; the holdout is purely temporal.

(2) *Seam state on the Gaussian scale.* The two columns of draw m's trajectory at $t \in \{T_o - 1, T_o\}$ are mapped *back* to the latent scale by the forward copula transforms,

$$\tau_{k,t}^{\star(m)} = \Phi^{-1}\big(G_{\alpha^{(m)}/2, \beta^{(m)}/2}(\tau_{k,t}^{(m)})\big), \qquad z_{k,t}^{\star(m)} = \Phi^{-1}\big(H_{\sigma_z}(z_{k,t}^{(m)})\big), \quad \sigma_z = (\tau_{k,t}^{(m)})^{-1/2}.$$

The state is taken from draw m's own imputed trajectory, *never* from the stationary law $\mathcal{N}(\mathbf{0}, \Sigma)$ – Corollary **??** is precisely the warning against the latter.

(3) *AR(2) recursion for $h = 1, \dots, H$.* On the Gaussian scale, with innovation variance fixed by Yule–Walker (**??**) so the marginal variance stays 1,

$$X^{\star(m)}_{k,T_o+h} = \phi_1^{(m)} X^{\star(m)}_{k,T_o+h-1} + \phi_2^{(m)} X^{\star(m)}_{k,T_o+h-2} + \sigma^{(m)}\,\xi, \quad \xi \sim \mathcal{N}(0,1),$$

applied independently to $\tau^\star$ and $z^\star$ with their own $\phi^{(m)}$. The i.i.d. baseline replaces this step by $X^{\star(m)}_{k,T_o+h} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ at every lead – the infinite-horizon limit (Corollary **??**).

(4) *Inverse copula back to observables.* Apply (**??**)–(**??**) to the forecasted latent slice; the τ–z coupling (**??**) is enforced *within* draw m, i.e. the half-normal scale σ_z uses the same draw's forecasted τ, not an independent marginal.

(5) *Spatial draw.* For each lead h (time $t = T_o + h$),

$$\hat{y}^{(m)}(s,t) = x(s,t)^\top \beta^{(m)} + \lambda^{(m)} z^{(m)}_{g(s,t),\,t} + \varepsilon^{(m)}(s,t), \qquad \varepsilon^{(m)}(\cdot,t) \sim \mathcal{N}\Big(\mathbf{0},\, (\tau^{(m)}_{g(s,t),\,t})^{-1} C^{(m)}\Big),$$

with $C^{(m)}$ the spatial correlation matrix at $(\rho^{(m)}, \nu^{(m)}, \gamma^{(m)})$.

Averaging $\hat{y}^{(m)}$ over $m = 1, \dots, M$ integrates parameter *and* seam-state uncertainty jointly (Proposition **??**); the per-block output is the predictive array of shape $M \times n_s \times H$. The full pseudocode is Algorithm **??**.

Non-negotiable design choices. The five stages above encode three constraints derived earlier and not negotiable in any reimplementation: (i) condition on the posterior of the seam state, not the stationary distribution (Corollary **??**); (ii) propagate full parameter *and* state uncertainty by iterating over posterior draws; and (iii) recurse on the latent Gaussian scale, applying the copula transforms (**??**)–(**??**) only after the recursion.

Proposition 5 (Coherence of Algorithm **??**)**.** *Algorithm **??** produces draws from the posterior predictive distribution of the time block, with both parameter uncertainty and seam-state uncertainty preserved, and with the τ–z coupling* (**??**) *respected.*

Proof. For each m, the tuple $\big(\phi^{(m)}, \sigma^{(m)}, \boldsymbol{X}^{(m)}_{T_o}, \beta^{(m)}, \lambda^{(m)}, \alpha^{(m)}, \beta^{(m)}, \rho^{(m)}, \nu^{(m)}, \gamma^{(m)}\big)$ is a single coherent sample from the joint posterior: the latent state $\boldsymbol{X}^{(m)}_{T_o}$ is an imputed quantity of draw m, not a fresh draw, so by Proposition **??** the recursion forecasts from the correct conditional law and, by Corollary **??**, does not collapse to the stationary marginal. Averaging the predictive over $m = 1, \dots, M$ integrates the posterior; no plug-in of posterior means occurs, so parameter uncertainty is retained. The τ–z coupling is enforced in line 12: the half-normal scale σ_z is evaluated at draw m's own forecasted $\tau^{(m)}_{\cdot,T_o+h}$, so τ and z are transformed jointly within the draw rather than from independent marginals. The AR innovations of $\tau^\star$ and $z^\star$ are a priori independent (Definition **??**), so no further cross-process innovation term is required; the posterior coupling is transmitted entirely through the shared seam state and shared parameters. □ □

Remark 4 (Implementation pitfalls). Four failure modes void Proposition **??** in practice. (a) *Seam from the stationary distribution* — redrawing $\boldsymbol{X}_{T_o}$ from $\mathcal{N}(\mathbf{0}, \Sigma)$ instead of extracting it from draw m; by Corollary **??** this erases the signal. (b) *Plug-in means* — forecasting from posterior means of ϕ and $\boldsymbol{X}_{T_o}$; this collapses uncertainty and under-disperses the predictive. (c) *Reseeding*

Algorithm 1 Posterior predictive forecast over a held-out time block

Input: MCMC draws $m = 1, \dots, M$; seam time T_o; horizon H; held-out sites $\mathcal{S}_{\text{val}}$.
Set a single RNG seed *once*, outside all loops.
for each draw $m = 1, \dots, M$ **do**
  for each latent process $X \in \{\tau^\star, z^\star, w^\star\}$ and knot k **do**
    Extract seam state $\big(X^{(m)}_{k,T_o-1}, X^{(m)}_{k,T_o}\big)$ directly from draw m's imputed latent trajectory.
    for $h = 1, \dots, H$ **do**
      $X^{(m)}_{k,T_o+h} \leftarrow \phi_1^{(m)} X^{(m)}_{k,T_o+h-1} + \phi_2^{(m)} X^{(m)}_{k,T_o+h-2} + \sigma^{(m)} \xi, \quad \xi \sim \mathcal{N}(0,1)$.
    end for
  end for
  for $h = 1, \dots, H$ **do**
    $\tau^{(m)}_{\cdot,T_o+h} \leftarrow G^{-1}_{\alpha^{(m)},\beta^{(m)}}\big(\Phi(\tau^{\star(m)}_{\cdot,T_o+h})\big)$.
    $z^{(m)}_{\cdot,T_o+h} \leftarrow H^{-1}_{\sigma_z}\big(\Phi(z^{\star(m)}_{\cdot,T_o+h})\big), \quad \sigma_z = (\tau^{(m)}_{\cdot,T_o+h})^{-1/2}$.
    $w^{(m)}_{\cdot,T_o+h} \leftarrow \Psi^{-1}\big(\Phi(w^{\star(m)}_{\cdot,T_o+h})\big)$; recompute membership g by (**??**).
    for each $s \in \mathcal{S}_{\text{val}}$ **do**
      $\mu \leftarrow x(s, T_o+h)^\top \beta^{(m)} + \lambda^{(m)} z^{(m)}_{g(s,T_o+h),T_o+h}$.
      Draw $y^{\star(m)}(s, T_o+h)$ from the spatial Gaussian field with mean μ and precision $\tau^{(m)}_{g(s,T_o+h),T_o+h}$.
    end for
  end for
end for
Output: predictive sample $\{y^{\star(m)}(s, T_o+h)\}_{m,s,h}$.

inside the loop — any `set.seed` call within the m- or h-loop breaks path coherence; the seed is set once, outside. (d) *Non-stationary draws* — posterior tail draws of ϕ may exit the triangle (**??**), making the recursion explode; either reparameterize to partial autocorrelations or reject such draws before forecasting. A further numerical hazard is Voronoi membership flipping: small drift in the forecasted $w_{k,t}$ can move a site across a knot boundary and discontinuously change μ. Averaging the predictive over many draws softens the partition; reporting a fixed-membership ablation ($g(s,t) \equiv g(s,T_o)$) alongside the full result isolates the AR(2) signal in τ and z from partition noise.

4.3 Lead-time scoring protocol

The energy and variogram scores of Definitions **??**–**??** are multivariate: each returns one number for an entire block vector and cannot be split by lead time. A lead-time curve therefore requires a *univariate* proper score evaluated separately at each h.

Definition 10 (Lead-time score curve)**.** Let $S\big(h; b, s\big)$ be a univariate proper score (e.g. CRPS, or the Brier score restricted to lead h) for the prediction at site s, lead time h, in block b. The lead-time curve is

$$\bar{S}(h) = \frac{1}{B\,|\mathcal{S}_{\text{val}}|} \sum_{b=1}^{B} \sum_{s \in \mathcal{S}_{\text{val}}} S\big(h; b, s\big), \qquad h = 1, \dots, H, \tag{18}$$

with a standard error at each h estimated from the dispersion of the B per-block means.

The evaluation reports two complementary objects.

(i) **Lead-time curve.** Plot $\bar{S}(h)$ from (**??**) for the AR(2) model and the i.i.d. baseline on common axes, with error bands. By Proposition **??** the AR(2) curve should lie below the baseline at small h and converge to it near $h^\star$. The crossing point is the headline quantity.

(ii) **Joint-structure summary.** Report the energy score (**??**) and variogram score (**??**) computed over the per-site length-H time vector, averaged over blocks and sites — one number per model that credits the temporal dependence the lead-time curve resolves.

Remark 5 (Verification checks). Three diagnostics confirm the forecast loop is correct. (1) With $H = 0$, re-scoring time T_o must give AR(2) and i.i.d. the same predictive — any difference is a seam-indexing bug. (2) A draw with $\phi = \mathbf{0}$ must collapse immediately to the marginal; residual AR-like behaviour indicates double-counted state. (3) A long horizon $H \gg h^\star$ with stationary ϕ must reproduce the $\mathcal{N}(0,1)$ latent marginal across draws, validating the innovation-variance normalization (**??**).

5 Conclusion and Future Work

The null result — AR(2) temporal pooling failing to beat an i.i.d. baseline on the Brier score — is fully explained by Propositions **??** and **??**: a marginal score under a spatial-only holdout makes the two models exchangeable, so the comparison degenerates to a bias–variance contest that the lower-dimensional model wins. The finding is a property of the experimental design, not evidence against temporal pooling.

The corrected framework of Section **??** removes the structural blindness. By holding out time blocks of length $H \lesssim h^\star$, forecasting from the posterior of the seam state (Algorithm **??**, Proposition **??**), and resolving performance by lead time (**??**), the evaluation exercises precisely the conditional forecast in which the AR(2) advantage resides (Proposition **??**). The expected signature of a genuine effect is a lead-time curve on which AR(2) beats the i.i.d. baseline at short horizons and converges to it near $h^\star$ (Proposition **??**).

Open items for the next iteration:

- **Block-length calibration.** Estimate ρ_* from the posterior of ϕ and set H from (**??**) per setting rather than by a fixed rule.
- **Partition uncertainty.** Compare the soft-membership predictive against the fixed-membership ablation of Remark **??**; report both so the AR(2) signal in τ, z is separated from Voronoi flipping noise.
- **Score sensitivity.** Contrast the lead-time CRPS curve with the energy and variogram summaries to confirm the conclusions are not an artifact of a single scoring rule.
- **Forecasting deployment.** Beyond block holdout, evaluate genuine forward forecasting from the end of the record, the setting in which temporal pooling is of direct operational value.

6 Implementation Audit Log

This section records the implementation defects encountered while operationalizing Section **??** into runnable code, the root cause of each, the patch applied, and the diagnostic of Remark **??** that did or should have caught it.

6.1 Incident A1: dimension drop in `fit$tau` when $K = 1$

Symptom. `forecast_block` halted with "`incorrect number of dimensions`" at the seam-state slice `tau_traj[m, , seam_cols]`.

Root cause. The MCMC backend returns the latent trajectory via `tau = keepers.tau[return.iters, , ]` without `drop = FALSE`. With `nknots = 1` the singleton knot axis is silently dropped, so `fit$tau` arrives as a 2-D matrix $[M, T_o]$ rather than the documented 3-D array $[M, K, T_o]$, and every index expression written for the 3-D shape breaks.

Fix. Restore the knot axis at the top of `forecast_block`: if `length(dim(tau_traj)) == 2`, wrap to `array(tau_traj, dim = c(M, 1, T_o))`. Extract the seam slice with `drop = FALSE` and reshape via `matrix(..., K, 2L)` so the $K = 1$ case yields a $K \times 2$ matrix (not a vector) when passed to `ar2_recurse`, whose body indexes `seam_state[, j]`.

Verification. The $H = 0$ parity diagnostic of Remark **??** (1) is designed to expose exactly this class of seam-indexing bug: a non-zero AR(2) versus i.i.d. score split at lead 0 is the signature.

6.2 Incident A2: $z \equiv 0$ under `temporal/ar2` updates

Symptom. Every cell of the AR(2) method's predictive sample $\hat{y}$ was `NaN` – $M \cdot n_s \cdot H \cdot B$ entries, zero finite values – and the NaN avalanche tripped `checkNumeric` inside `scoringRules::crps_sample`. The i.i.d. method (which uses a closed-form Gibbs update for z, bypassing the copula) was unaffected, which is what isolated the failure to the AR(2) branch.

Root cause. The half-normal copula transform applied to the latent skewness process is

$$z^{\star}_{k,t} = \Phi^{-1}\big(H_{\sigma_z}(z_{k,t})\big), \qquad H_{\sigma_z}(0) = 0 \implies \Phi^{-1}(0) = -\infty.$$

The backend default `z.init = 0` therefore feeds $z^{\star} = -\infty$ to the very first iteration of `updateZTS_AR2`. The Metropolis log-ratio then evaluates the Gaussian transition kernel `ar2_conditional_log_density` at $-\infty$, where its exponent reduces to $\infty - \infty =$ NaN. The acceptance step is guarded by `if (!is.na(R))` in `update_params_ar2.R`, which silently *skips* every proposal when R is `NaN` – so the chain never leaves $z \equiv 0$ and $\phi_z \equiv \mathbf{0}$ for any iteration, while λ becomes unidentified ($\lambda \cdot z = 0$ regardless of λ) and its posterior wanders over $[-47, 38]$ despite the true $\lambda = 3$.

In `forecast_block` the seam state is then $z^{\star}_{T_o} = -\infty$, and the AR(2) recursion produces

$$\phi_{z,1} \cdot (-\infty) + \phi_{z,2} \cdot (-\infty) = 0 \cdot (-\infty) = \text{NaN}$$

because $\phi_z = \mathbf{0}$ exactly; the `NaN` cascades through the inverse copula and the spatial draw into every cell of $\hat{y}$.

Fix. Pass `z.init = 0.01` explicitly to `mcmc()` in `run-settings.R`. The first iteration's `hn.cop` then evaluates finitely, the MH ratio is real-valued, the chain mixes, and the imputed z, ϕ_z, and λ trajectories become non-degenerate.

Backend follow-up. The silent acceptance guard is the deeper defect: a `NaN` log-ratio is not a probabilistic rejection but a numerical failure, and conflating the two suppresses the diagnostic signal that matters most. A defensive patch in `updateZTS_AR2` should either count `NaN` as a rejection *with* a `warning("MH ratio NaN; chain may be stuck")`, or refuse `z.init <= 0` when `temporalz = TRUE` – so the user sees the failure rather than absorbing it.

Verification. The $\phi = \mathbf{0}$ collapse diagnostic of Remark **??** (2) is the canonical check, but only if the imputed z trajectory is inspected *before* forecasting: `range(fit$z)` stuck at $[0, 0]$ is the signature, and `Var(fit$lambda)` of order 10^2 alongside truth $\lambda = 3$ is the corroborating tell. Both can be checked in one line per fit and added to the post-fit health screen of Remark **??**.

6.3 Incident A3: `parLapply` missing `tasks` export

Symptom. The driver halted with "`object 'tasks' not found`" inside the PSOCK workers whenever `workers > 1`; the single-worker branch ran cleanly with the same input.

Root cause. The `parLapply` body references `tasks$method[i]` and `tasks$dataset[i]`, but `tasks` was missing from the `clusterExport` call. PSOCK workers serialize the function with its closure, but free-name lookups on the worker happen against the worker's global environment, not the closure – so the symbol is unbound there. The single-worker branch evaluates the body in the script's own environment, which masks the bug entirely.

Fix. Add `"tasks"` to the `clusterExport` list. Every free name the worker body references must be either passed as a function argument or exported explicitly.

Verification. A two-worker, two-task smoke run before scaling out exposes this class of bug in seconds; serial-only verification cannot.

6.4 Lessons

- **Silent acceptance is the worst kind of failure.** A Metropolis update that returns `NaN` must either fail loudly or count as a rejection *with* a warning, but never be hidden by an `is.na` gate; Incident A2 was undetectable at the fit stage and produced a structurally well-formed output array.

- **Singleton dimensions are platform-specific landmines.** R drops them by default; passing `drop = FALSE` or restoring the axis explicitly is the only portable way to write K-agnostic forecast code (Incident A1).

- **Test the parallel path on a small grid.** PSOCK scoping bugs are invisible in serial mode (Incident A3); a two-worker, two-task run before fanning out is the cheapest possible insurance.

- **Augment Remark ?? with a pre-forecast fit audit.** Confirm Var(fit$z) > 0, Var(fit$phi.z) > 0, and range(fit$lambda) concentrated near the prior mode *before* invoking `forecast_block`. A zero-variance z trajectory is a sufficient (and very cheap) diagnostic for Incident A2.

References

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